

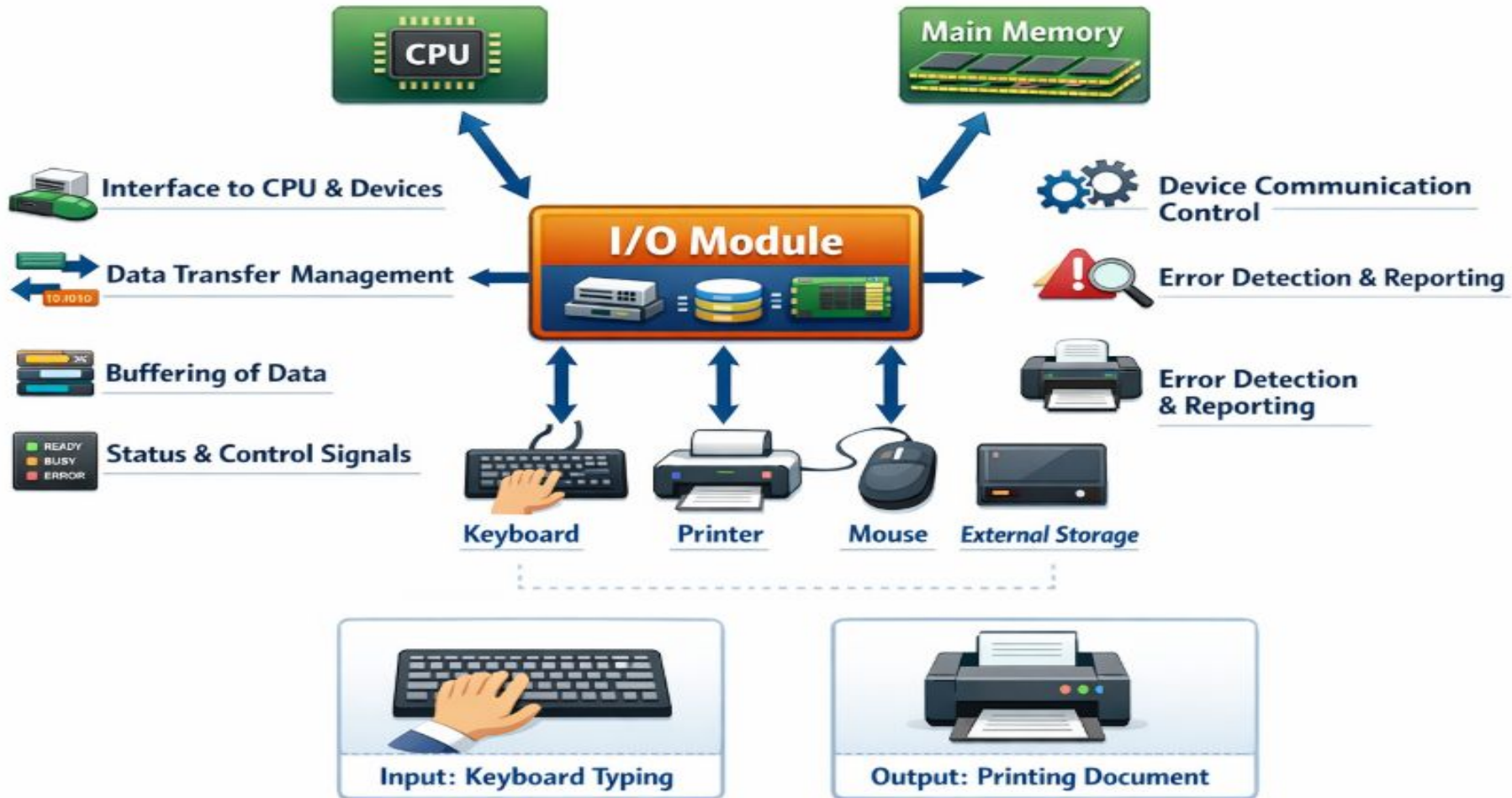


Input/Output

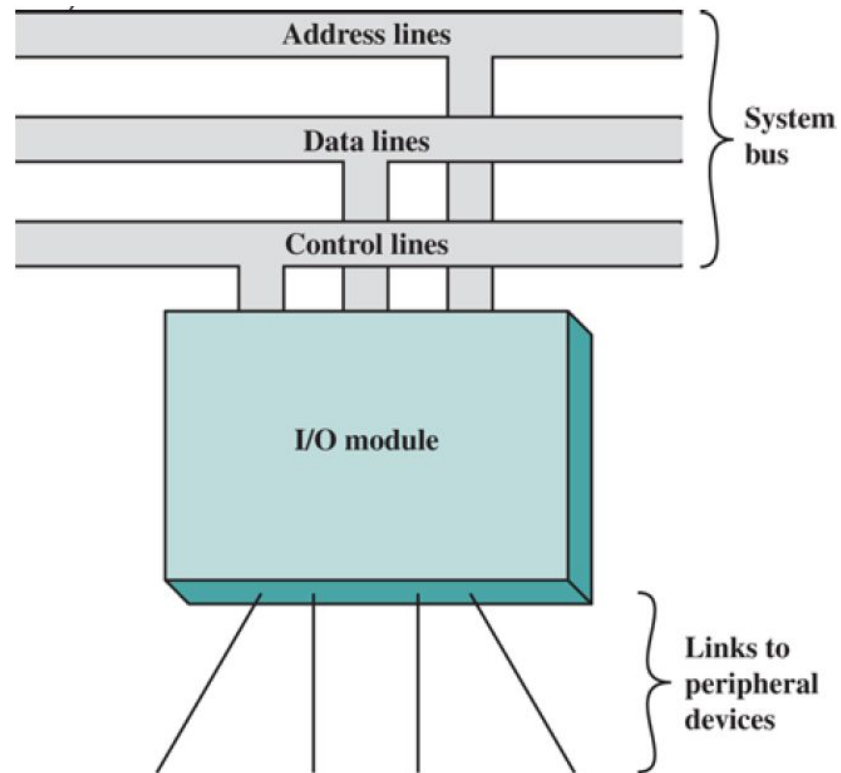
Definition of I/O Module

- ❖ an Input–Output (I/O) module is a hardware component that manages and controls the communication between the computer's central processing unit (CPU), main memory, and external peripheral devices.
- ❖ acts as an interface that allows data to be transferred between the computer system and input/output devices such as keyboards, printers, sensors, and storage devices.

Uses of the I/O Module in Computer Organization & Architecture



Generic Model of an I/O Module Interface



- ✓ Interface to the processor and memory via the system bus or central switch.
- ✓ Interface to one or more peripheral devices by tailored data links.

External Devices

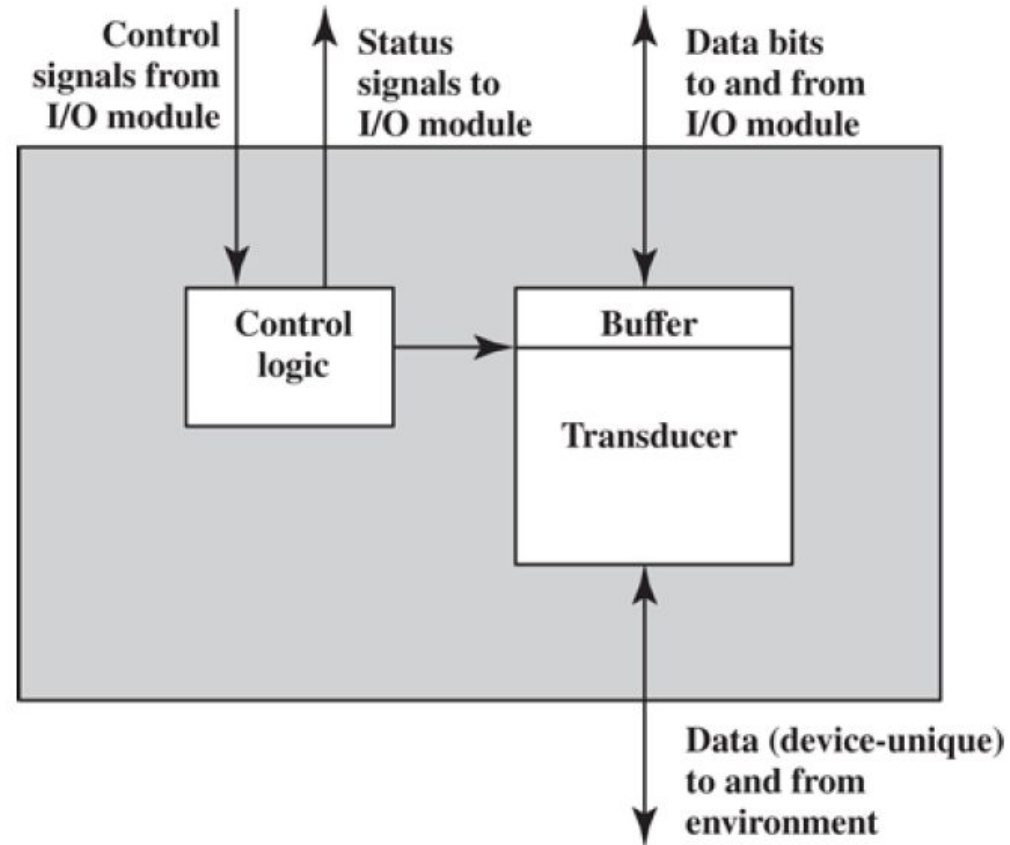
- I/O operations are accomplished through a wide assortment of external devices that provide a means of **exchanging data between the external environment and the computer**.
- An external device attaches to the computer by a **link to an I/O module**.
- The link is used to exchange **control, status, and data** between the I/O module and the external device.
- An external device connected to an I/O module is often referred to as a **peripheral device** or, simply, a peripheral

External Devices

We can broadly classify external devices into three categories:

1. **Human readable**: Suitable for communicating with the computer user;
2. **Machine readable**: Suitable for communicating with equipment;
3. **Communication**: Suitable for communicating with remote devices..
 - a) Examples of human- readable devices are **video display terminals (VDTs) and printers**.
 - b) Examples of machine- readable devices **are magnetic disk and tape systems, and sensors** and actuators, such as are used in a robotics application.
 - c) Communication devices allow a computer to exchange data with a remote device, which may be a human- readable device, such as a terminal, a machine- readable device, or even another computer.

Block Diagram of an External Device



Block Diagram of an External Device

- ✓ The interface to the I/O module is in the form of **control, data, and status signals**.
- ✓ **Control signals** determine the function that the device will perform, such as send data to the I/O module (INPUT or READ), accept data from the I/O module (OUTPUT or WRITE), report status, or perform some control function particular to the device (e.g., position a disk head).
- ✓ **Data** are in the form of a **set of bits** to be sent to or received from the I/O module.
- ✓ **Status** signals indicate the state of the device. Examples are **READY/NOT-READY** to show whether the device is ready for data transfer.

Block Diagram of an External Device

- ✓ **Control logic** associated with the device **controls the device's operation** in response to direction from the I/O module.
- ✓ The **transducer** **converts data from electrical to other forms** of energy during output and from other forms to electrical during input.
- ✓ Typically, a **buffer** is associated with the transducer to **temporarily hold data** being transferred between the I/O module and the external environment.
- ✓ A buffer size **of 8 to 16 bits** is common for serial devices,
 1. Keyboard/Monitor
 2. Disk Drive

Keyboard/Monitor

- ✓ The most common means of computer/user interaction is a **keyboard/monitor** arrangement. The user provides input through the keyboard, the input is then transmitted to the computer and may also be displayed on the monitor. In addition, the monitor displays data provided by the computer.
- ✓ The basic unit of exchange is the **character**. Associated with each character is a code, **typically 7 or 8 bits in length**. The most commonly used text code is the **International Reference Alphabet (IRA)**.
- ✓ Each character in this code is represented by a unique 7-bit binary code; thus, 128 different characters can be represented.
- ✓ Characters are of two types: **printable and control**.
- ✓ Printable characters are the **alphabetic, numeric, and special characters** that can be printed on paper or displayed on a screen.
- ✓ Some of the control characters have to do with **controlling the printing or displaying of characters**; an example is carriage return.

Keyboard/Monitor

- ✓ For keyboard input, when the user depresses a key, this generates an electronic signal that is interpreted by the transducer in the keyboard and translated into the bit pattern of the corresponding IRA code. This bit pattern is then transmitted to the I/O module in the computer. At the computer, the text can be stored in the same IRA code. On output, IRA code characters are transmitted to an external device from the I/O module.
- ✓ The transducer at the device interprets this code and sends the required electronic signals to the output device either to display the indicated character or perform the requested control function.

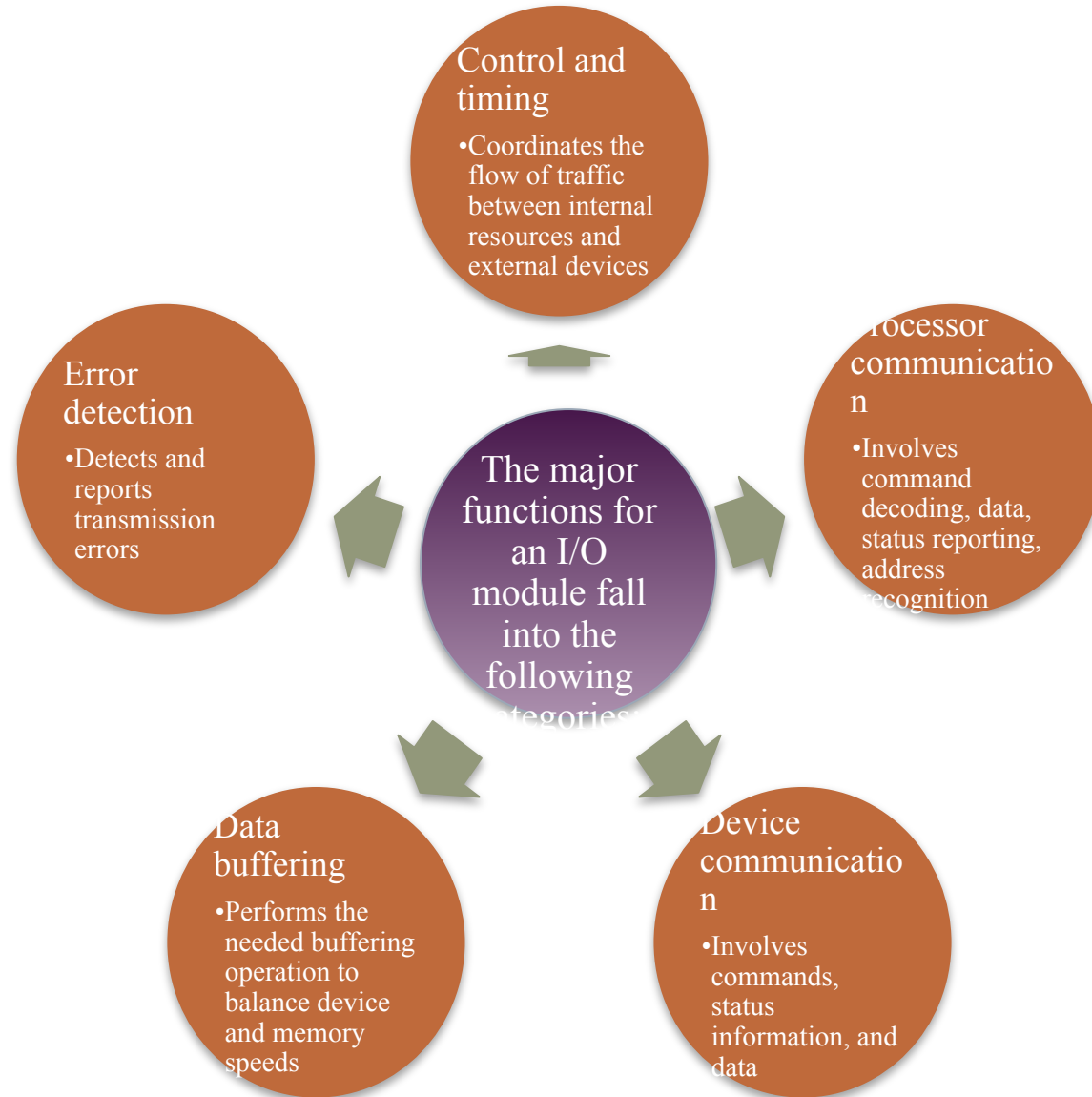
Disk Drive

- ✓ A disk drive contains electronics for exchanging data, control, and status signals with an I/O module plus the electronics for controlling the disk read/write mechanism. In a fixed-head disk, the transducer is capable of converting between the magnetic patterns on the moving disk surface and bits in the device's buffer.



I/O Modules

Module Function



Module Function

- ✓ The major functions or requirements for an I/O module fall into the following categories:
 1. Control and timing
 2. Processor communication
 3. Device communication
 4. Data buffering
 5. Error detection
- ✓ During any period of time, the processor may communicate with one or more external devices in unpredictable patterns, depending on the program's need for I/O.

Control and Timing

- ✓ Thus, the I/O function includes a **control and timing** requirement, to coordinate the flow of traffic between internal resources and external devices.
- ✓ For example, the control of the transfer of data from an external device to the processor might involve the following sequence of steps:
 1. The processor interrogates the I/O module to check the status of the attached device.
 2. The I/O module returns the device status.
 3. If the device is operational and ready to transmit, the processor requests the transfer of data, by means of a command to the I/O module.
 4. The I/O module obtains a unit of data (e.g., 8 or 16 bits) from the external device.
 5. The data are transferred from the I/O module to the processor.

Processor communication

- ✓ The preceding simplified scenario also illustrates that the I/O module must communicate with the processor and with the external device. Processor communication involves the following:

Command decoding: The I/O module accepts commands from the processor, typically sent as signals on the control bus. For example, an I/O module for a disk drive might accept the following commands: READ SECTOR, WRITE SECTOR, SEEK track number, and SCAN record ID.

Data: Data are exchanged between the processor and the I/O module over the data bus.

Status reporting: Because peripherals are so slow, it is important to know the status of the I/O module. For example, if an I/O module is asked to send data to the processor (read), it may not be ready to do so because it is still working on the previous I/O command. This fact can be reported with a status signal. Common status signals are BUSY and READY. There may also be signals to report various error conditions.

Address recognition: Just as each word of memory has an address, so does each I/O device.

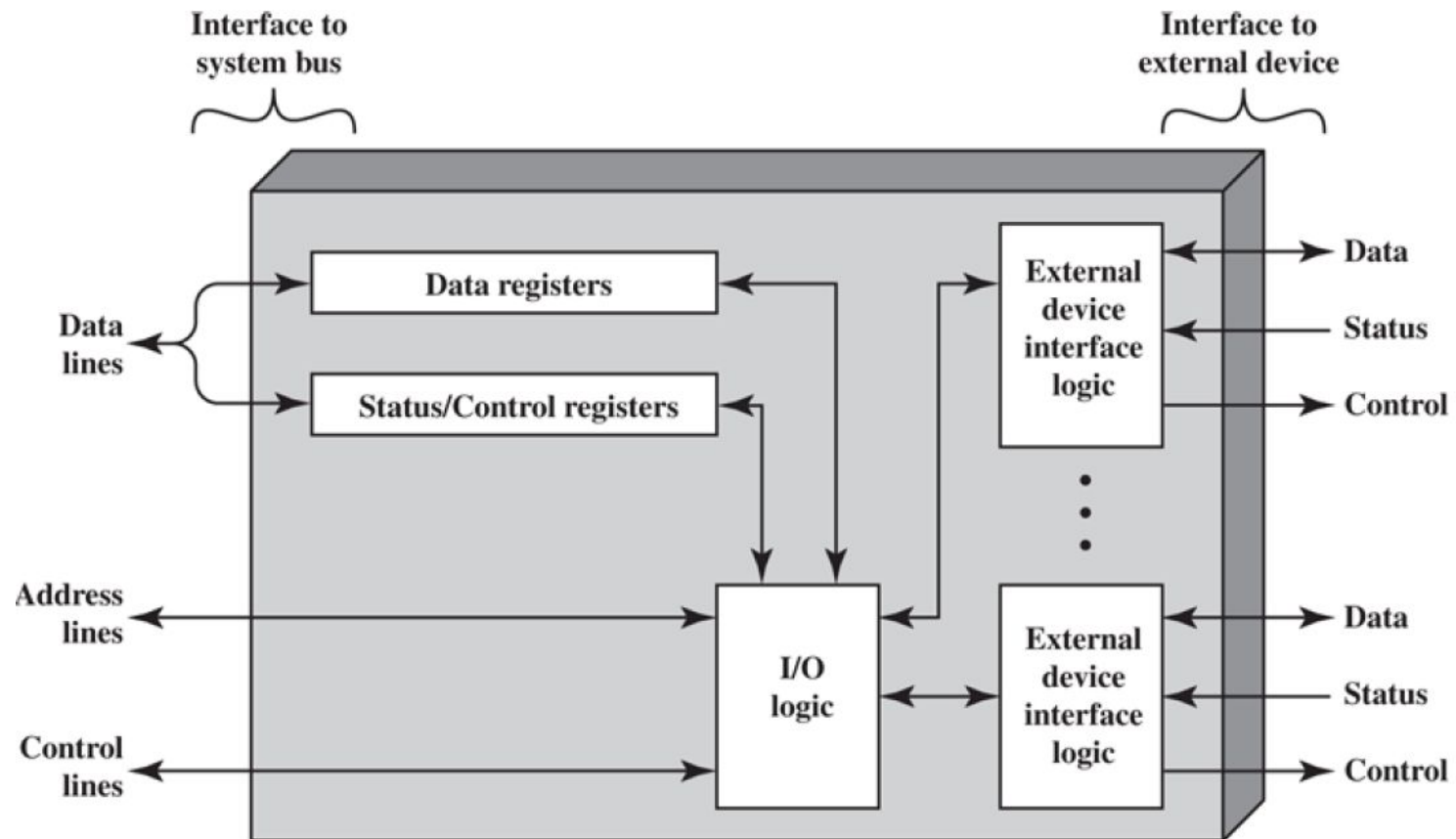
Module Function

- ✓ On the other side, the I/O module must be able to perform **device communication**. This communication involves commands, status information, and data.
- ✓ An essential task of an I/O module is **data buffering**. The need for this function is apparent from Whereas the transfer rate into and out of main memory or the processor is quite high, the rate is orders of magnitude lower for many peripheral devices and covers a wide range.
- ✓ Data coming from main memory are sent to an I/O module in a rapid burst.
- ✓ The data are buffered in the I/O module and then sent to the peripheral device at its data rate.
- ✓ In the opposite direction, data are buffered so as not to tie up the memory in a slow transfer operation.
- ✓ Thus, the I/O module must be able to operate at both device and memory speeds.
- ✓ Similarly, if the I/O device operates at a rate higher than the memory access rate, then the I/O module performs the needed buffering operation.

Error detection

- ✓ Finally, an I/O module is often responsible for **error detection** and for subsequently reporting errors to the processor.
- ✓ One class of errors includes **mechanical and electrical malfunctions** reported by the device (e.g., **paper jam, bad disk track**).
- ✓ Another class consists of unintentional changes to the **bit pattern** as it is transmitted from device to I/O module. Some form of error- detecting code is often used to detect transmission errors. A simple example is the use of a parity bit on each character of data.
- ✓ For example, the IRA character code occupies 7 bits of a byte. The eighth bit is set so that the total number of 1s in the byte is even (even parity) or odd (odd parity).
- ✓ When a byte is received, the I/O module checks the parity to determine whether an error has occurred.

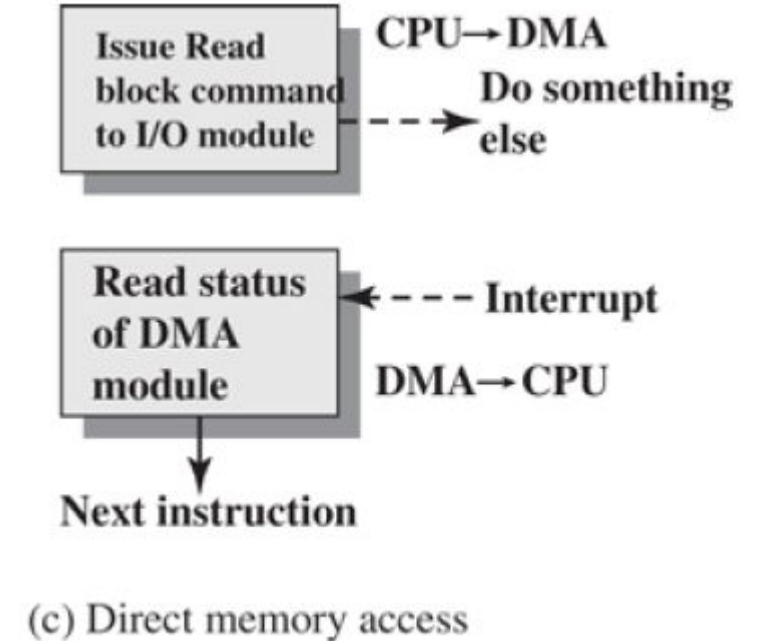
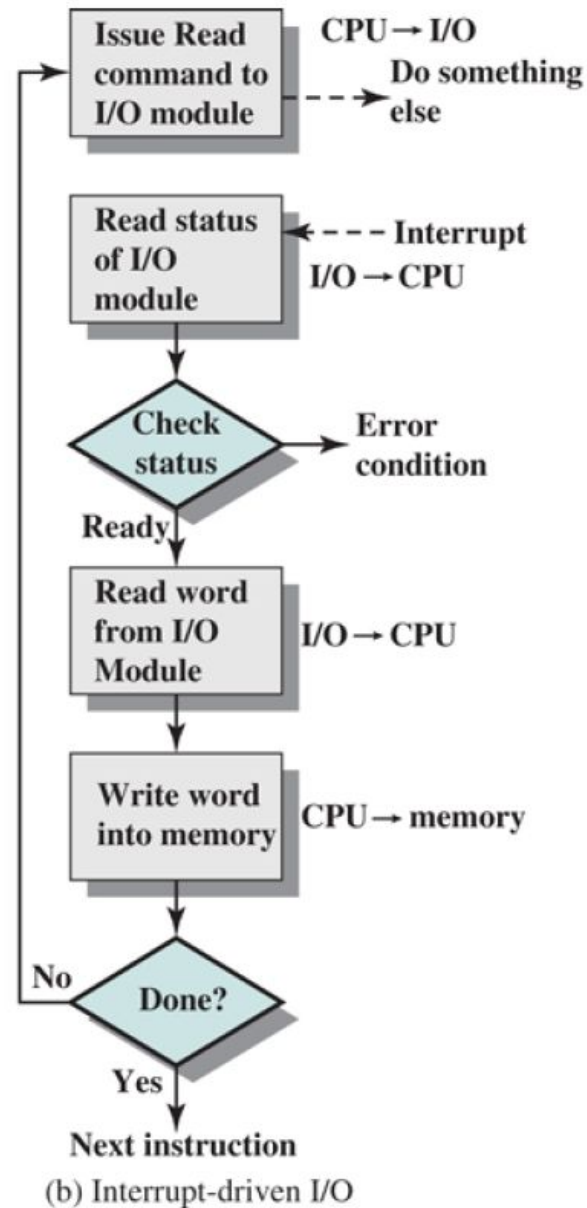
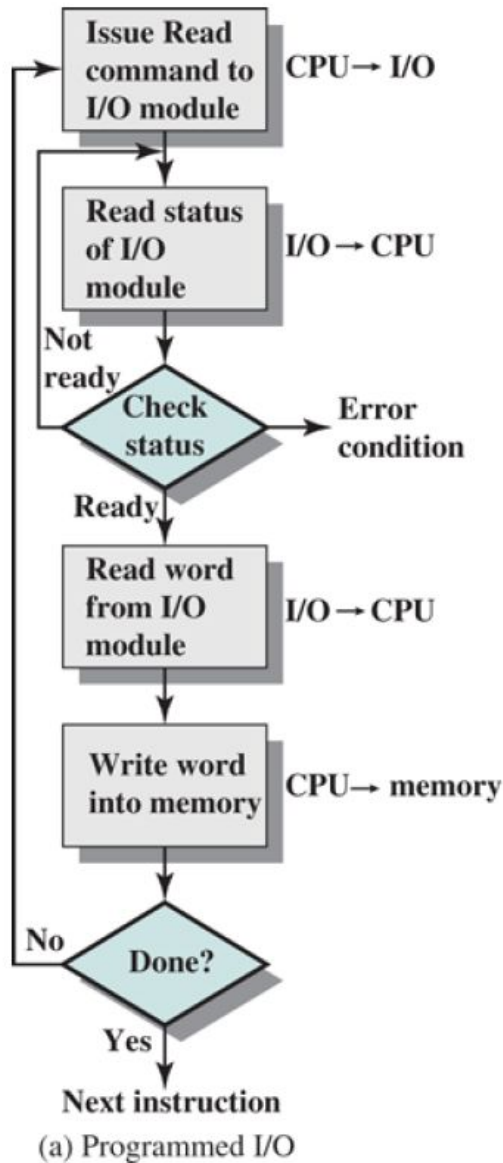
I/O Module Structure



I/O Module Structure

- ✓ The module connects to the rest of the computer through a set of signal lines (e.g., system bus lines).
- ✓ Data transferred to and from the module are buffered in one or more data registers.
- ✓ A status register may also function as a control register, to accept detailed control information from the processor.
- ✓ The logic within the module interacts with the processor via a set of control lines.
- ✓ The processor uses the control lines to issue commands to the I/O module. Some of the control lines may be used by the I/O module (e.g., for arbitration and status signals). The module must also be able to recognize and generate addresses associated with the devices it controls.
- ✓ Each I/O module has a unique address or, if it controls more than one external device, a unique set of addresses.
- ✓ Finally, the I/O module contains logic specific to the interface with each device that it controls.

Three Techniques for Input of a Block of Data



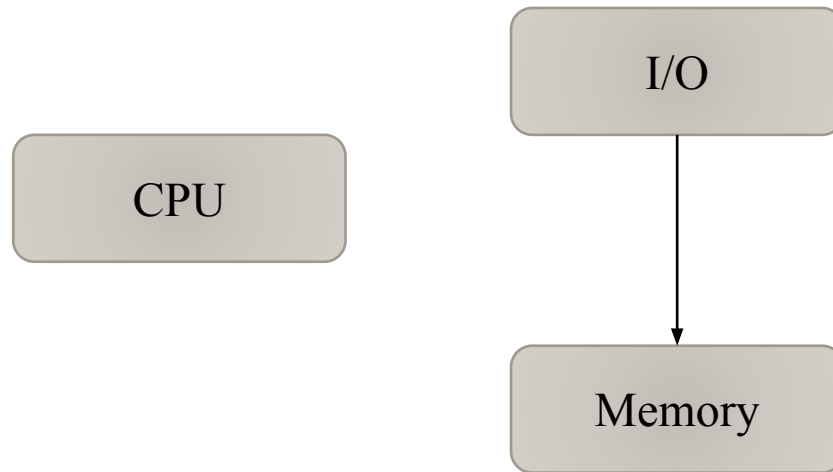


Direct Memory Access (8257)

Drawbacks of Programmed and Interrupt-Driven I/O

1. The I/O transfer rate is limited by the speed with which the processor can test and service a device.
 2. The processor is tied up in managing an I/O transfer; a number of instructions must be executed for each I/O transfer
- ✓ There is somewhat of a trade-off between these two drawbacks. Consider the transfer of a block of data. Using simple programmed I/O, the processor is dedicated to the task of I/O and can move data at a rather high rate, at the cost of doing nothing else. Interrupt I/O frees up the processor to some extent at the expense of the I/O transfer rate. Nevertheless, both methods have an adverse impact on both processor activity and I/O transfer rate.

DMA



Direct Memory Access

DMA Controller is a hardware device that allows I/O devices to directly access memory with less participation of the processor

What is a DMA Controller?

- (1) DMA uses hardware for accessing the memory, that hardware is called a DMA Controller.
- (2) It has the work of transferring the data between Input Output devices and main memory with very less interaction with the processor.
- (3) The direct Memory Access Controller is a control unit, which has the work of transferring data.

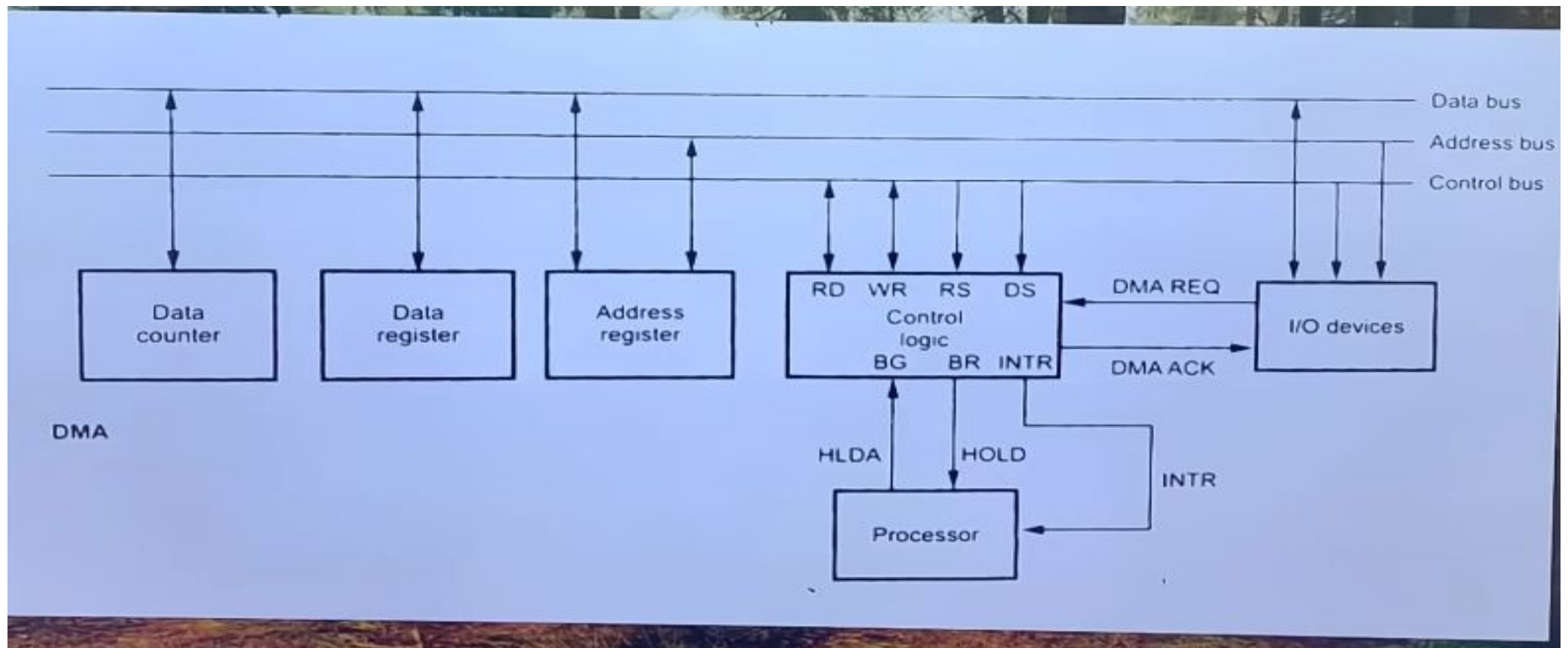
DMA operations

- ✓ It is designed by Intel to transfer data at the fastest rate.
- ✓ It allows the device to transfer data directly to or from memory without CPU involvement.

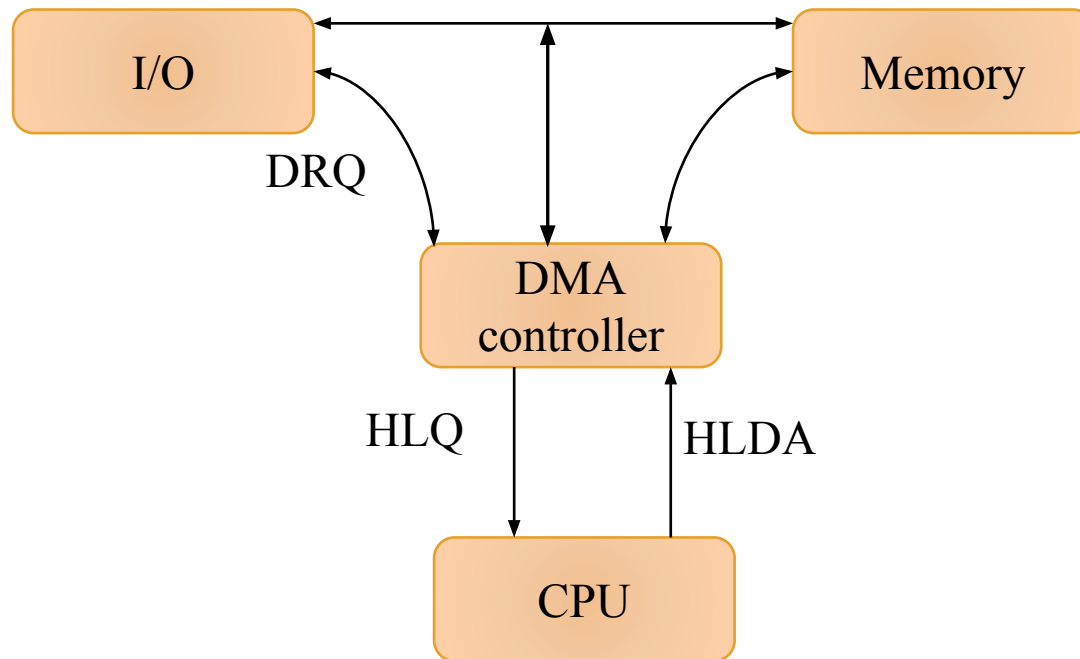
DMA operations:

1. Device wants to send data to memory
 - a) DMA request DRQ to DMA controller
 - b) DMA controller send HLQ(hold request) to CPU and waits for the CPU HLDA(hold acknowledgement)
 - c) The CPU leaves the control over bus and Ack the HLDA signal to DMA controller.
 - d) The CPU is in HOLD state. DMA controller has to manage operations over buses between CPU, memory and I/O devices.

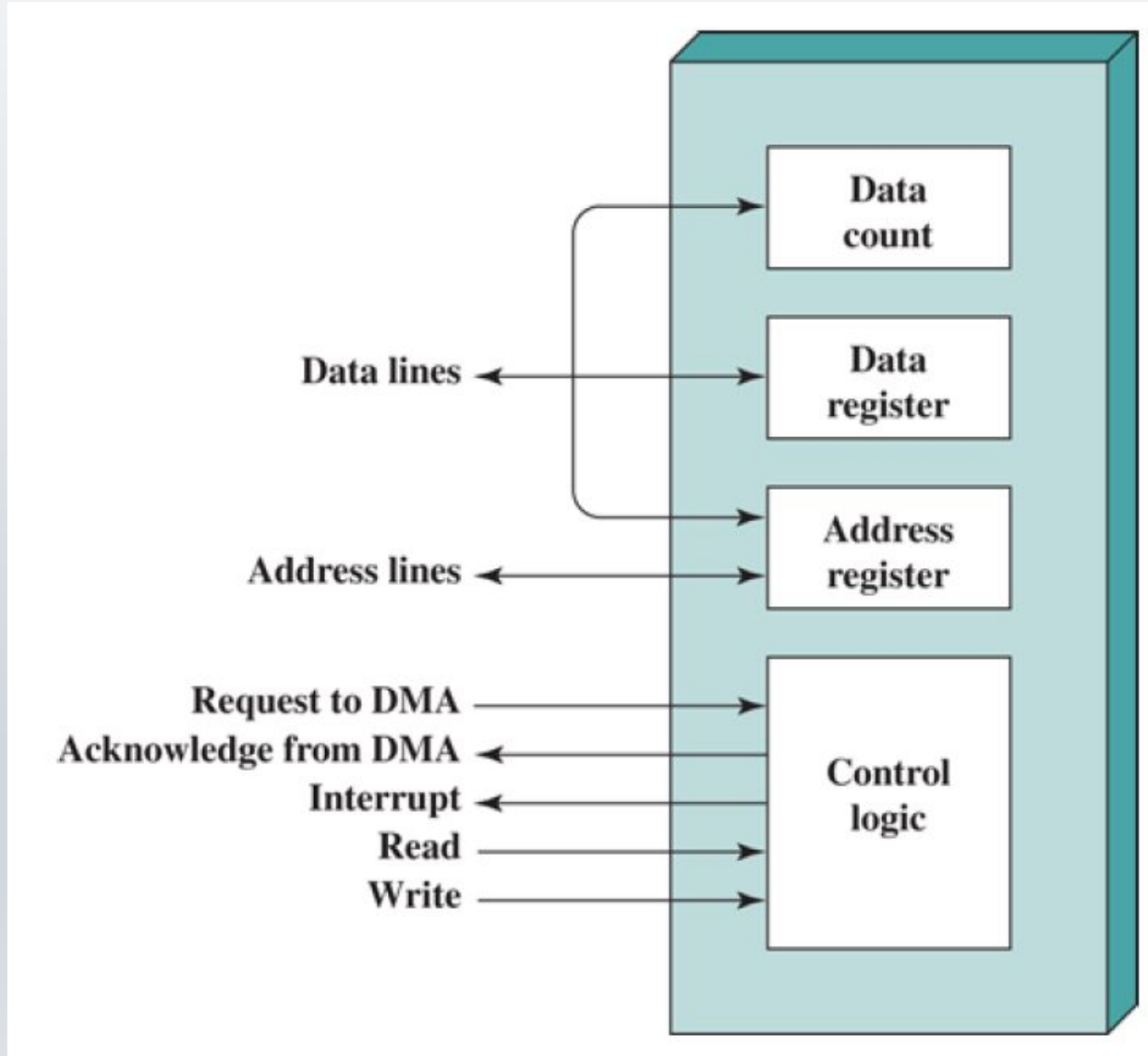
Block Diagram



DMA operations



DMA Block Diagram

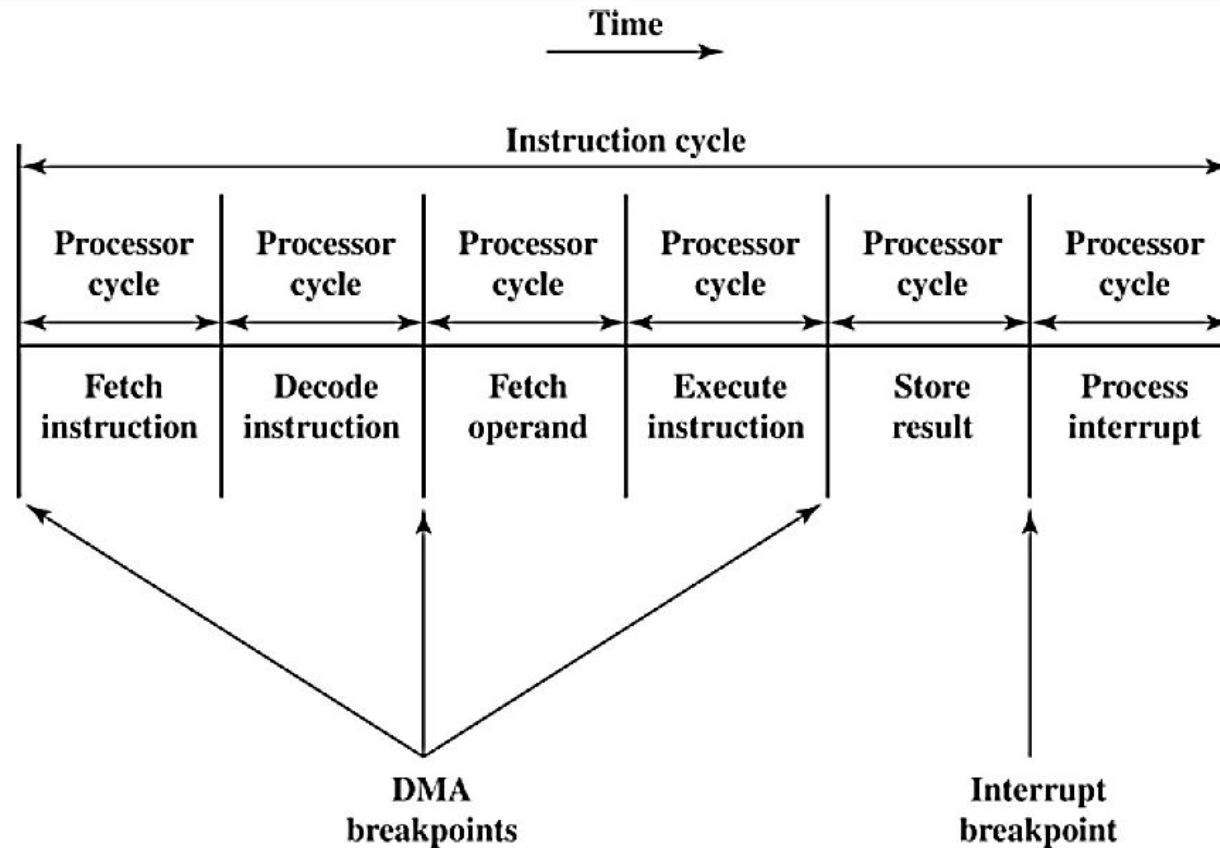


DMA operations

When the processor wishes to read or write a block of data, it issues a command to the DMA module, by sending to the DMA module the following information:

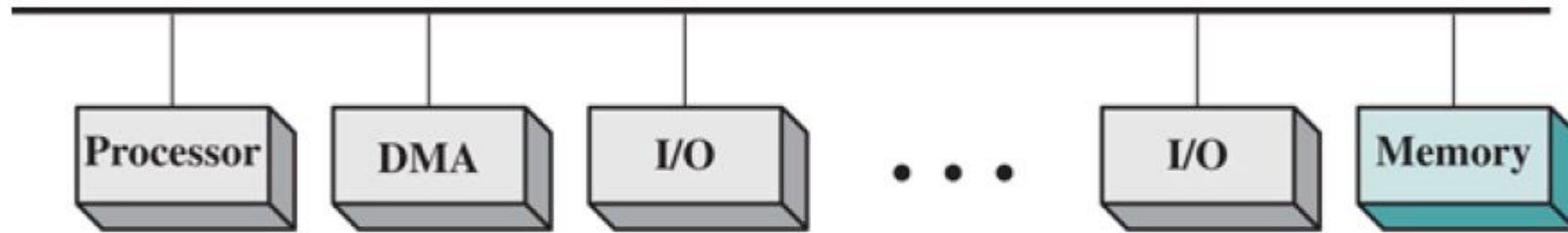
- ✓ Whether a read or write is requested, using the read or write control line between the processor and the DMA module.
- ✓ The address of the I/O device involved, communicated on the data lines.
- ✓ The starting location in memory to read from or write to, communicated on the data lines and stored by the DMA module in its address register.
- ✓ The number of words to be read or written, again communicated via the data lines and stored in the data count register.

DMA and Interrupt Breakpoints during an Instruction Cycle



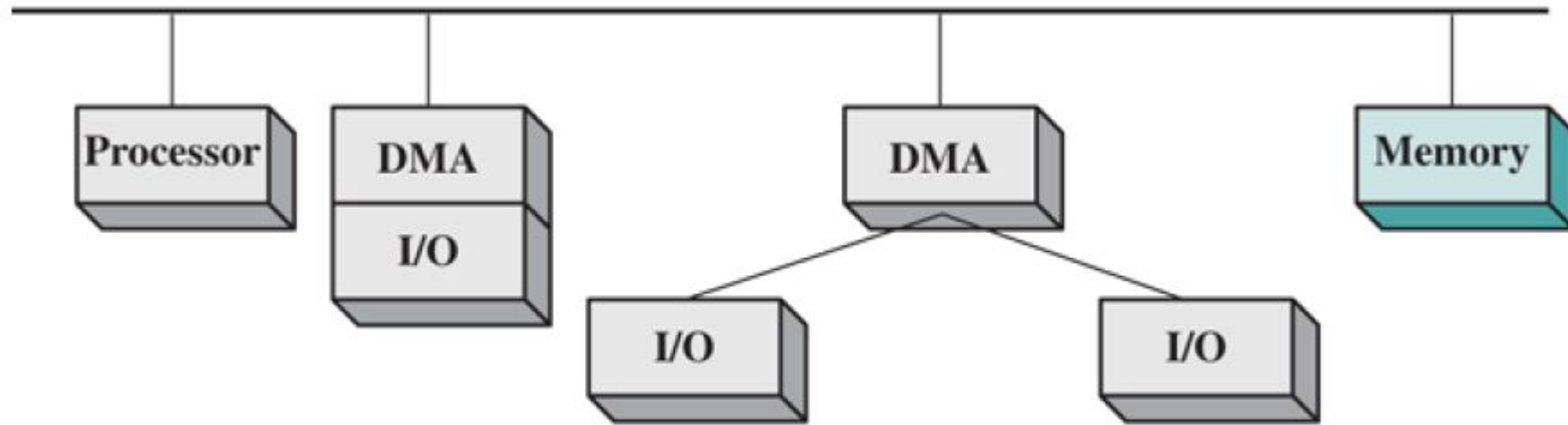
- (1) DMA consists of data count, data register, address register and control logic.
- (2) Data counter register stores the number which gives the number data transfers to be done in one DMA cycle.
- (3) It is automatically decremented after each word transfer.
- (4) Data register acts as buffer whereas address register initially holds the starting address of the device.
- (5) Actually, it stores the address of the next word to be transferred. It is automatically incremented or decremented after each word transfer.
- (6) After each transfer, data counter is tested for zero.
- (7) When the data count reaches zero, the DMA transfer halts.
- (8) The DMA controller is normally provided with an interrupts capability, in which case it sends an interrupt to processor to signal the end of the I/O data transfer.

Single bus without DMA



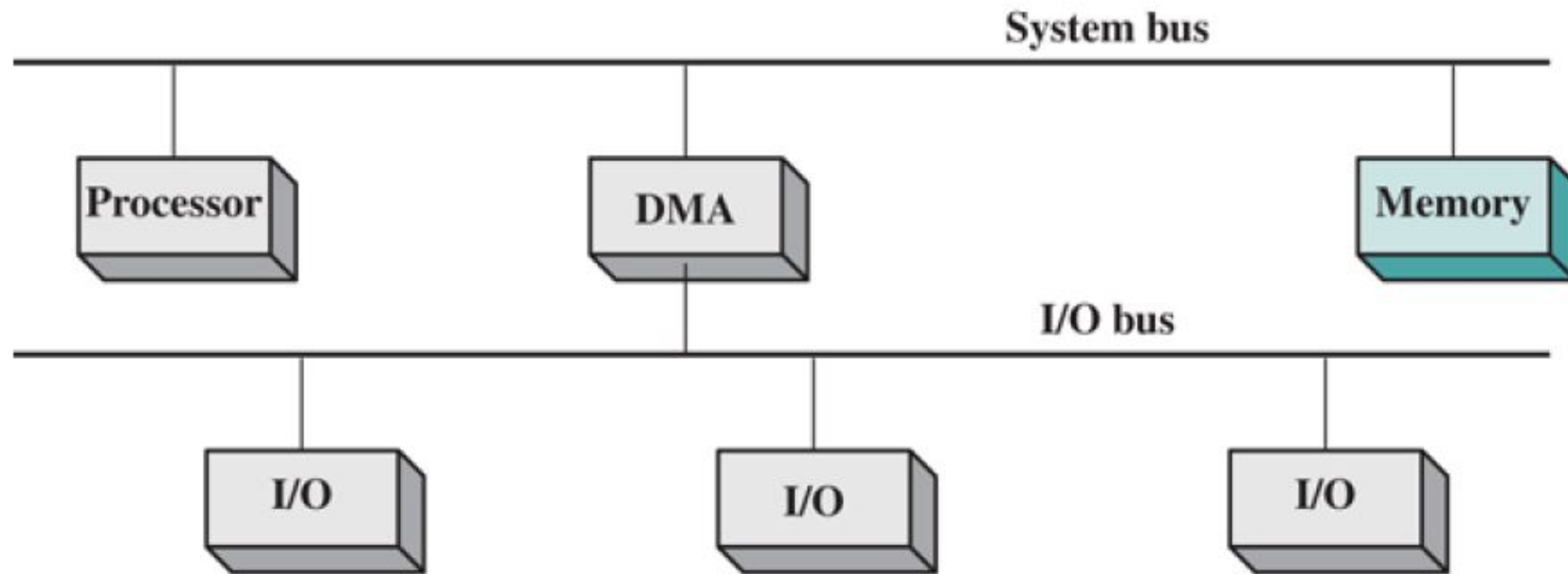
(a) Single-bus, detached DMA

Single bus integrated DMA-I/O



(b) Single-bus, integrated DMA-I/O

I/O bus



(c) I/O bus